

Experiment-6

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Course Code: -MLA0305

Slot-B

Title

Implementation of Monte Carlo Prediction and Control for Reinforcement Learning

Aim

To implement **Monte Carlo Prediction and Monte Carlo Control** methods for Reinforcement Learning, estimate the value function from complete episodes, and learn an **optimal policy** using an ϵ -greedy strategy.

Procedure

1. Define a simple Reinforcement Learning environment with a finite set of states and actions.
2. Define the reward function and terminal state.
3. Select the discount factor and number of training episodes.
4. Generate complete episodes by allowing the agent to interact with the environment.
5. Calculate the **return** by processing the rewards obtained during an episode.
6. Implement **First-Visit Monte Carlo Prediction**:
 - Identify the first occurrence of each state.
 - Calculate its return.
 - Update the estimated state value .
7. Implement **Monte Carlo Control** using an ϵ -greedy policy.
8. Initialize the action-value function .
9. Generate episodes using exploration and exploitation.
10. Calculate the return for each state-action pair.
11. Update using the observed returns.
12. Improve the policy by selecting the action with the maximum Q-value:
$$\pi(s) = \operatorname{argmax} Q(s, a)$$
13. Repeat the process for the specified number of episodes.
14. Calculate the optimal policy from the learned Q-values.
15. Evaluate the performance using reward, return, success rate, and episode length.

16. Generate graphs showing value convergence, reward, Q-values, and exploration vs exploitation.

17. Display summary tables and export the results to CSV files.

Important Equations

Monte Carlo Return

$$G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \gamma^3 R_{t+3} + \dots$$

State Value

$$V(s) = \text{Average of returns observed after visiting } s$$

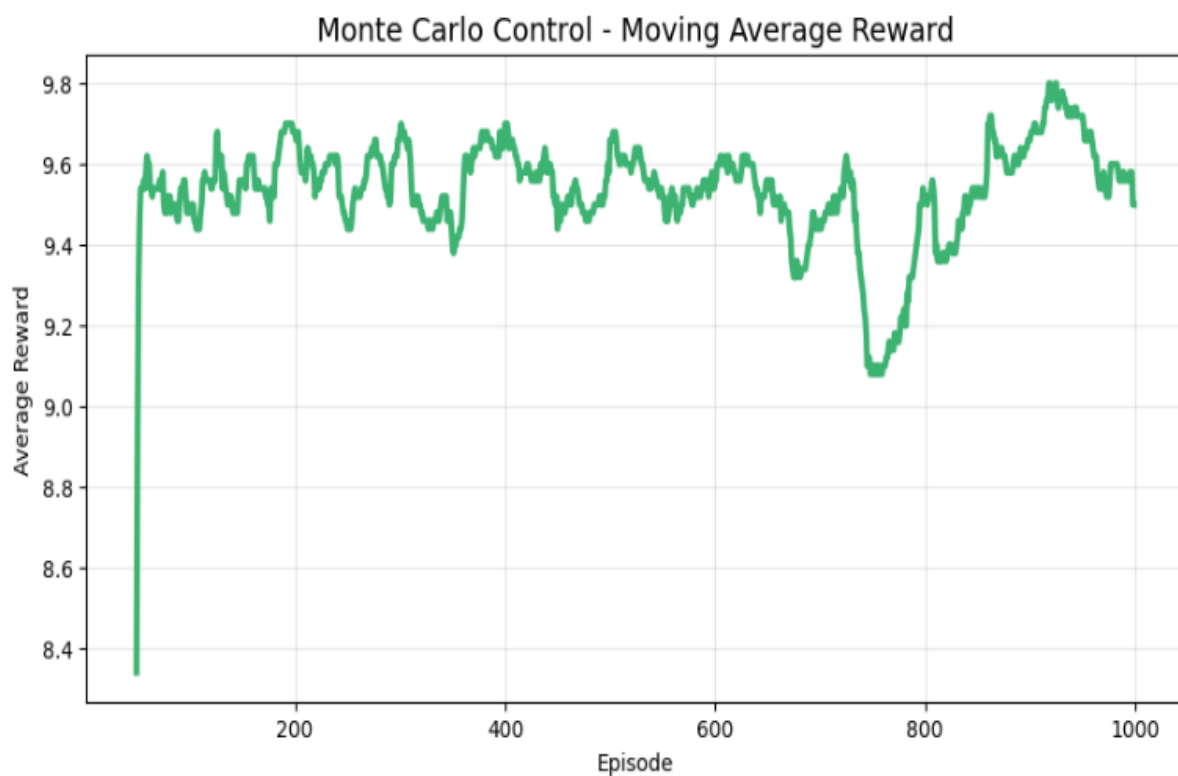
Action Value

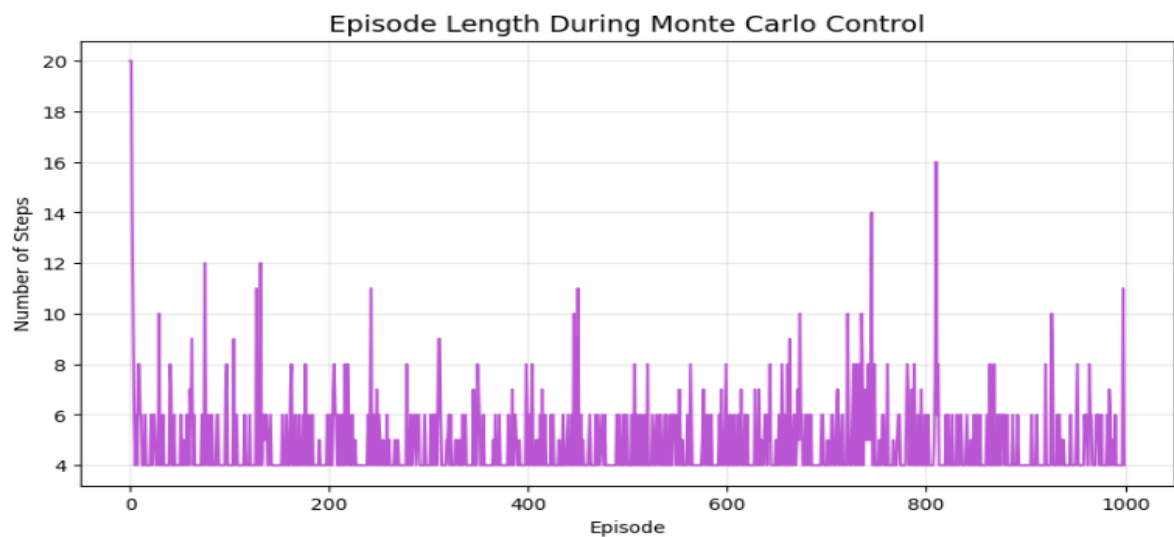
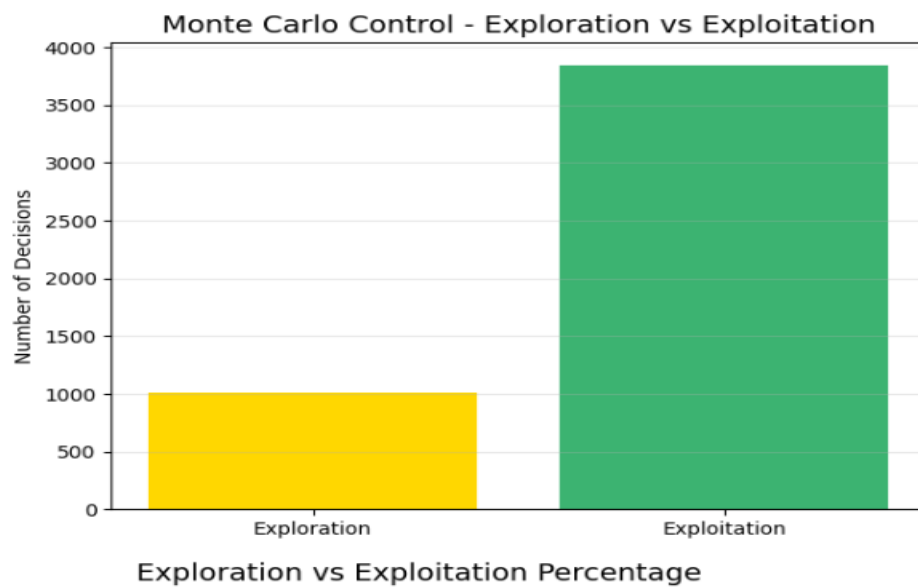
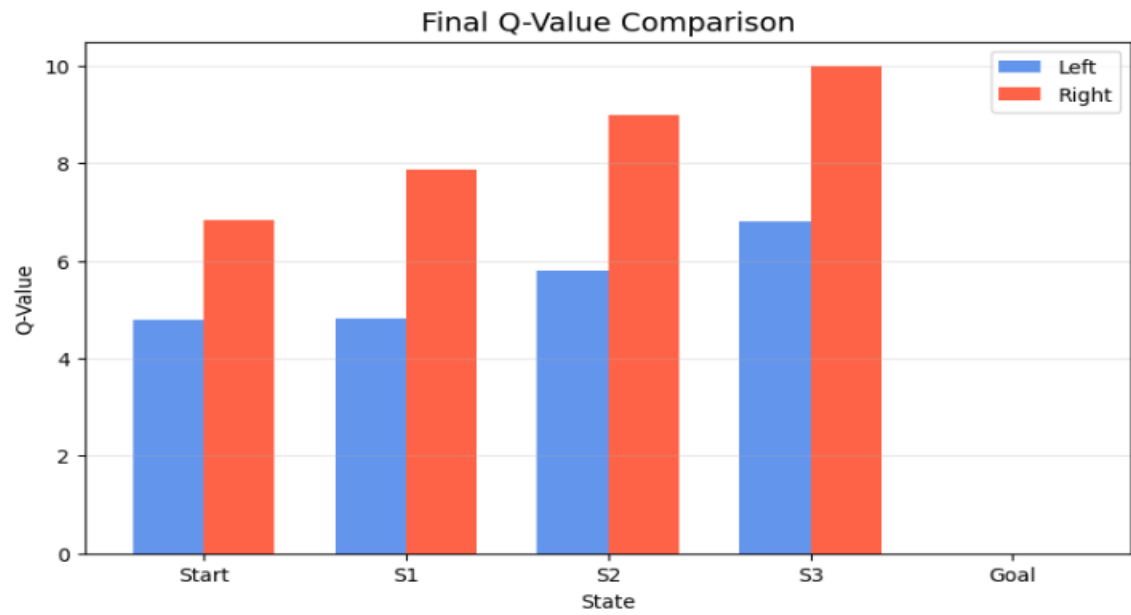
$$Q(s, a) = \text{Average of returns observed after taking } a \text{ in } s$$

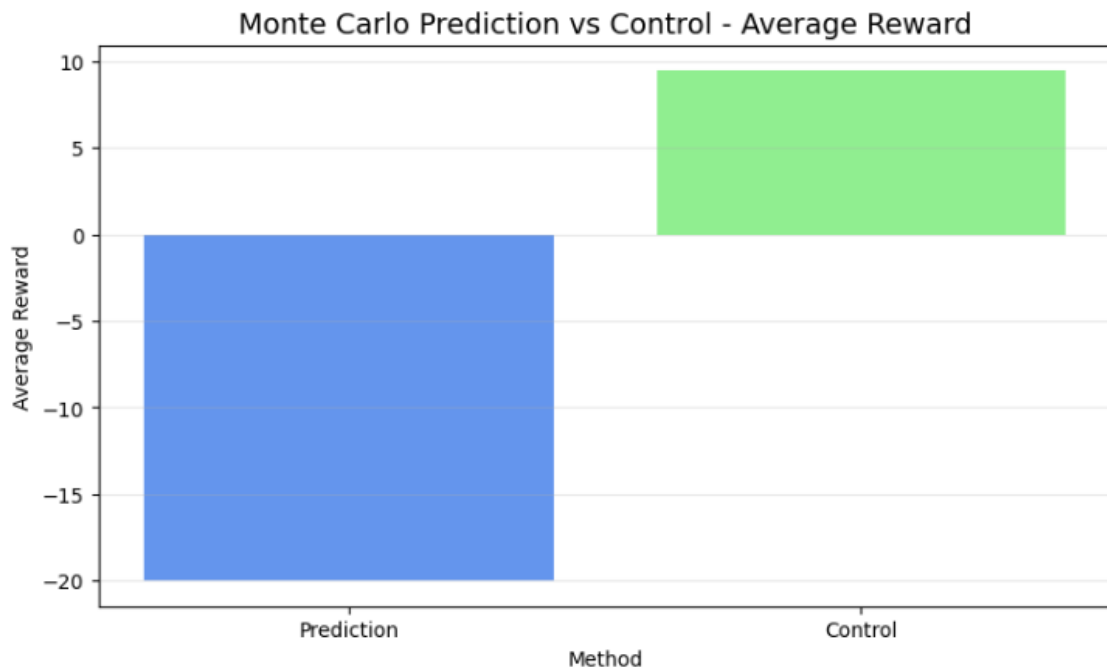
Optimal Policy

$$\pi^*(s) = \operatorname{argmax}_a Q(s, a)$$

Visualizations







Summary Table – Monte Carlo Prediction

State	Estimated Value	Number of Visits
Start	Obtained from program	Obtained
S1	Obtained from program	Obtained
S2	Obtained from program	Obtained
S3	Obtained from program	Obtained
Goal	Obtained from program	Obtained

Summary Table – Monte Carlo Control

State	Left Q-Value	Right Q-Value	Optimal Action
Start	Obtained	Obtained	Obtained
S1	Obtained	Obtained	Obtained
S2	Obtained	Obtained	Obtained
S3	Obtained	Obtained	Obtained
Goal	Obtained	Obtained	Terminal

Result

Monte Carlo Prediction and Control were successfully implemented.